

Dominic DeCarlo

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EDUCATION

University of Pittsburgh

Pittsburgh, PA

Bachelor of Science in Computer Science and Data Science

August 2023 – April 2027

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Software Engineering, Data Science, Systems Software, Discrete Math, Calculus, Linear Algebra, Stats, Probability

EXPERIENCE

AI & Automation Intern

May 2026 – December 2026

Thermo Fisher Scientific

Pittsburgh, PA

- Engineered an ensemble ML model predicting customer experience scores and root-cause risk factors for IT incident tickets, saving managers 98% of manual triage time and projecting ~\$110k in operational savings.
- Assessed frontier model vulnerabilities and developed AI security frameworks to enable the safe deployment of high-performing LLMs.

IT Intern

May 2025 – August 2025

National Philanthropic Trust

Philadelphia, PA

- Built a Python-based AI/ML model to forecast contribution trends, helping the team proactively strategize for the high-volume giving season.
- Performed data analysis on complex datasets to develop business-focused data visualizations (Sankey charts, Power BI dashboards), improving reporting efficiency and providing actionable insights into donor behavior.

PROJECTS

Built From Scratch Large Language Model (LLM) | *Python, PyTorch, CUDA*

- Architected a custom transformer-based neural network from the ground up, executing distributed pre-training pipelines across high-performance A100/H100 GPU clusters to optimize the model architecture.
- Fine-tuned conversational behavior using a curated proprietary dataset and engineered a stable deployment pipeline to serve the optimized model weights via a live, web-hosted interface.

AI Video Security Platform (1st Place, SteelHacks XIII) | *Next.js, LoRA, Multimodal AI*

- Engineered an automated surveillance system leveraging custom trained video encoders for rapid anomaly screening paired with multimodal VLMs to analyze incidents across continuous CCTV footage in real time.
- Developed a full-stack operator workspace featuring automated risk scoring, timestamped incident timelines, and multi-channel audio dispatch via speech synthesis APIs.

AI Soccer Analysis System | *Python, OpenCV, YOLO*

- Developed an AI/ML system using Python and computer vision to analyze soccer games and provide insights.
- Implemented real-time object detection to analyze player movement, important game events, etc.

Fine-Tuned Local AI Coding & Voice Assistant | *PyTorch, Unsloth, QLoRA*

- Fine-tuned a 9B parameter LLM via QLoRA, improving ARC complex reasoning benchmark scores by +13.3% and surpassing the baseline 27B parameter model across 6 of 7 standard NLP evaluation benchmarks.
- Deployed a fast, local voice agent using STT & TTS pipelines with an LM Studio backend for live hands-free use.

Real-Time Prediction Market Arbitrage SaaS | *Python, FastAPI, Google Stitch*

- Engineered a Python scanner to instantly detect profitable arbitrage bets across Kalshi and Polymarket.
- Built an interactive dashboard to sort live market spreads by profit margin with direct execution links to exchanges.

CAMPUS INVOLVEMENT

Pitt Computer Science Club

August 2024 – Present

- Contributed to collaborative projects and technical discussions, boosting problem-solving and technical knowledge.

Panther Equity

January 2025 – Present

- Gained hands-on finance experience through market trend analysis and equity research for simulated portfolios.

TECHNICAL SKILLS

Languages: Python, C++, R, SQL, Java, Go, HTML/CSS, JavaScript, TypeScript, Bash, X86, MIPS

Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, NumPy, React, Svelte, Node.js, OpenCV

Developer Tools: GitHub, Git, n8n, StackStorm, Power Automate, Power BI, Markdown, Stitch, Excel